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Agency Name:	National Science Foundation
Agency Tracking Number:	2502380
Organization:	Division Of Mathematical Sciences
NSF Program:	ALGEBRA,NUMBER THEORY,AND COM
PI/PD:	Bruce, Juliette
Application Title:	Tropicalizations of Shimura Varieties

Review 1

Rating:

Multiple Rating: (Excellent/Very Good)

Review:

Summary

Proposal Synopsis

The PI proposes to study the tropicalization of Shimura varieties using techniques from homological algebra, algebraic geometry, and combinatorics. The broader impacts include continu organization of several conference series, with an emphasis on inclusivity, as well as advocacy through SLMath-hosted AWM retreats.

Intellectual Merit

Strengths

Rating: Very good

Based on previous work with collaborators, the PI outlines an ambitious plan to extend results on the cohomology of Shimura varieties, which parameterize objects like elliptic curves or al varieties. Several conjectures are presented with detailed approaches. In particular, the PI breaks down 4 major directions including:

1. Developing a theory of tropicalizations of Shimura varieties.
2. Giving a modular interpretation of the tropicalizations.
3. Compute weight zero cohomologies of the associated spaces, perhaps with computer software.
4. Explore additional algebraic structures on these cohomologies.

The PI and collaborators have several publications and in-progress projects working toward the stated goals of this proposal.

The PI outlines a couple projects suitable for student advisees.

Weaknesses

Many of the technical details are omitted for readability, but that means the proposed results and their consequences are at times difficult to judge.

Some of the conjectures seem a bit speculative in full generality but the PI has a good track record for producing such results.

Broader Impacts*Strengths*

Rating: Excellent

The PI has a strong record of organizing conferences and meetings, especially for women and LGBTQ+ mathematicians. PI proposes to continue this work including a satellite conference 2026 ICM, expanding the GEMS workshops for gender equity in math, and a yearly retreat at SLMath for women to promote advocacy for women in mathematics. The PI serves on the S Committee on Women in Mathematics and also plans to organize a summer REU for LGBTQ+ mathematicians. The PI expects to begin mentoring students soon.

Weaknesses

The PI does not have graduate advisees but that is understandable.

Additional Solicitation-specific Review Criteria*Strengths & Weaknesses*

During their most recent funding period, the PI organized more than 10 national and international conferences. They served on the Executive Committee of AWM, as a board member for Spectra, and the SLMath Committee on Women in Mathematics. The PI supervised several undergraduate research projects, participated in Vakil's Algebraic Geometry in the Time to Co project. They published 8 new papers and preprints, which have appeared in top journals such as Algebra and Number Theory, Geometry and Topology, and Journal of Algebra.

Summary Statement

Overall, this is an impressive and ambitious proposal. The outlined research would greatly generalize work on certain moduli spaces. The outlined broader impacts are likely to have a strong positive influence on inclusivity in the math research community.

This proposal falls in the middle third of ones I reviewed.

Review 2**Rating:**

Multiple Rating: (Excellent/Very Good)

Review:**Summary****Proposal Synopsis**

The broad goal of the project is to study Shimura varieties and in particular to understand their cohomology. To achieve this goal the PI proposes to develop the theory of tropicalization of Shimura varieties and use the combinatorics of that space to compute cohomology.

Intellectual Merit*Strengths*

Rating: V

This is a very well-written proposal. The PI has a great record of past research accomplishments. The proposal outlines ways each of the proposed problems will be addressed. First proposed project: The research program is clearly outlined after stating the research problem (Problem 2.6). This is work in progress and seems the PI has made significant progress already in order to make this conjecture. Second suggested project: this seems to be a generalization of previous results of the PI. The third part of the project: is computational, based on adapting code the co-authors used for previous work.

Weaknesses

The projects proposed seem to be work in progress or generalizations of past work.

Broader Impacts*Strengths*

Rating: E

The PI has an impressive list of events organized for someone this early in their career. The PI has immediate plans for conference organization for the upcoming year(s). The PI also has to support mathematicians from historically underrepresented backgrounds through work with SLMath and Dartmouth.

Weaknesses

The ideas for the REU and mentorship programs as well as the SLMath workshop are not very fleshed out. The PI is already in the process of mentoring students.

Additional Solicitation-specific Review Criteria

Strengths & Weaknesses

Rating: E

The PI has held NSF Graduate Research Fellowship and NSF Postdoctoral Research Fellowship: The results are total of 13 papers (some published in very prestigious journals) and huge involvement in organizing and mentoring activities.

Summary Statement

this is a good proposal with a strong chance of success

Review 3

Rating:

Good

Review:

Summary

Proposal Synopsis

The PI proposes to study the tropical geometry of Shimura varieties, and to continue her successful record of conference organization and mentorship.

Intellectual Merit

Strengths

-The proposed project on Shimura varieties is a very natural line of inquiry, and the PI clearly has the technical chops to carry it out, given her impressive results on the weight zero cohomology of the moduli space of abelian varieties.

-The PI is clearly a very strong mathematician with a broad background and impressive publication record.

-I find question 5.5 especially interesting and unique due to the potential connections with number theory.

-Intellectual merit rating: G

Weaknesses

-The proposal is one-dimensional: besides for the discussion of prior work, the proposal is entirely focused on the weight-zero cohomology of Shimura varieties. This is certainly an interesting problem deserving of funding, but I would have liked to see some discussion of other directions the PI would like to go in as well. To give one example, the connection to number theory in question 5.5 could have been fleshed out more.

-The weight-zero direction continues a program initiated by the PI's postdoc advisor and their collaborators. There is nothing at all wrong with this, but the proposal under review does not include the development of an individual research program, unlike several other proposals under review.

Broader Impacts

Strengths

-The PI's prior organizational work supporting women and gender minorities in mathematics is very inspirational, and it merits financial support.

-In particular, the high level of effort she puts into organizing inclusive conferences is extremely admirable and should be emulated by many more mathematicians. Conferences are one of the main ways that mathematicians meet each other and discuss their ideas. It is extremely important that they are spaces where mathematicians of all identities feel included and safe to share ideas and ask questions. Otherwise, conferences can be discouraging to early career mathematicians from underrepresented groups.

-The PI's idea to start a mentorship program for LGBTQ+ students applying to graduate school is a good one, and will help increase participation of underrepresented groups in mathematics.

-The PI hopes to bring small groups of women and underrepresented minorities in mathematics together at SLMath to brainstorm ways to increase equity in mathematics. Such retreats se a good incubators for potential broader impacts.

-Broader impacts rating: V/G

Weaknesses

-The proposal is lacking in specific details about activities that the grant might support. I think the PI has a lot of excellent ideas for future broader impacts, but they are not fully fleshed out proposal. Here is one example: the idea for an REU program specifically for LGBTQ+ students is an excellent one that certainly should be pursued, but almost no details are provided abo steps the PI would take to make this happen (e.g. who else the PI is working with on making this happen, how the PI would find potential mentors, how a location would be chosen, obtaini funding, etc.).

-Let me emphasize that I do not at all doubt that the PI could make these activities happen, but some other proposals I am reviewing have provided a lot more detail about how proposed t impacts activities will be carried out.

Additional Solicitation-specific Review Criteria
Strengths & Weaknesses

As for results from prior NSF support (postdoctoral fellowship), I rate the intellectual merit to be V and broader impacts to be E.

Summary Statement

The proposed direction on the tropical geometry of Shimura varieties is very interesting, and the PI has a great record of both research and service that I am confident she will continue. Hl since the research focus is narrow compared to some other proposals I am reviewing, and some of the broader impacts of the proposal are lacking in specificity, I can only rate this propos 'Good'. The PI is a promising, talented early-career researcher whom I admire, but I do not feel that this proposal is completely reflective of her excellence.

Of the 7 proposals for standard grants I have reviewed[Material redacted per NSF Policy], this one is ranked 7th.

Of all of the proposals I have reviewed[Material redacted per NSF Policy], this one is ranked 9th.

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